



University of Science, VNU-HCM
Faculty of Information Technology
Honors Program

VOICE-BASED KEY GENERATION FOR USER AUTHENTICATION USING SAMPLE-THEN-LOCK CONSTRUCTION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

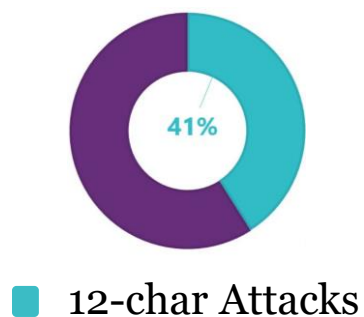
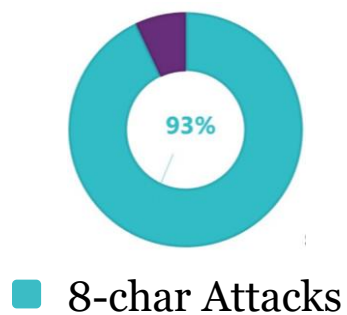
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Outline

- 1 Existing Authentication Methods
- 2 Solution & Objectives
- 3 Approaches & Architecture
- 4 Experimental Result & Security
- 5 Conclusion & Future Works

Password - The Weakest Link



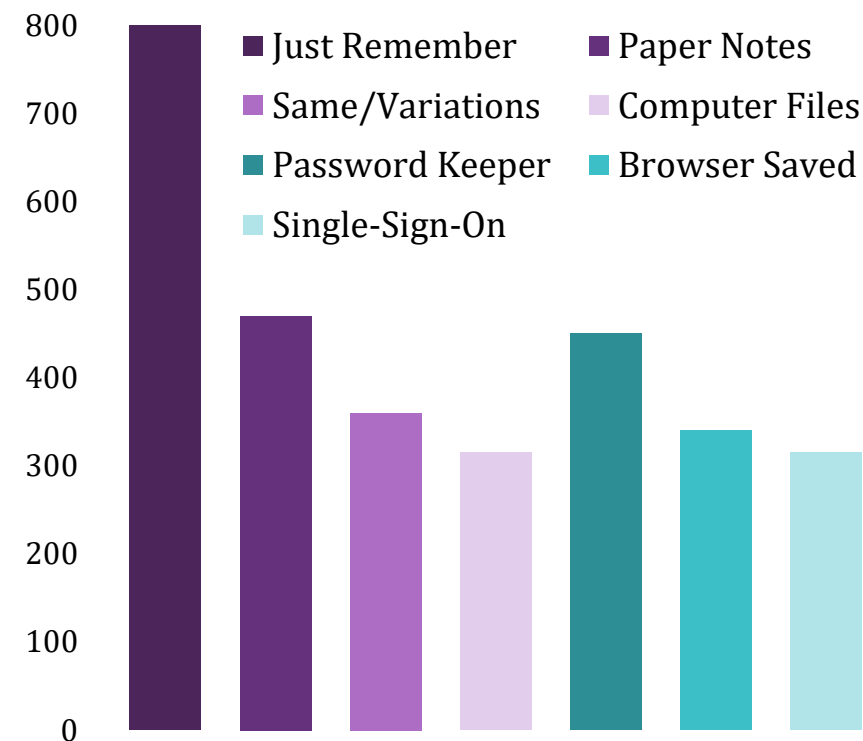
Top 10 passwords

used in real brute force attacks
with at least 12 characters

- | | |
|-------------------------------|-----------------------|
| 1 ^_ ^\$\$wanniMaBl:: 1433 vl | 6 P@ssw0rd5tgb |
| 2 almalinux8svm | 7 adminbigdata |
| 3 dbname=template0 | 8 Pa\$\$w0rdp!@# |
| 4 shabixuege!@# | 9 adm1nistrator1 |
| 5 @\$\$W0rd0123 | 10 administrator!@#\$ |

On a survey of 2000 people for having **11 or more** passwords:

- For working – 48%
- For personal – 71%



65% reuse passwords by **just remembering**
54% rely on **insecure methods** for managing

Biometric Authentication

Advantages:

- No need to remember complicated strings.
- Forging or stealing biometric is more difficult.

Disadvantages:

- Permanent loss if a template is compromised.
- Cryptographic challenging due to float values.
- Not applicable for data encryption.



Voice Recognition



Ear Shape
Recognition



DNA Matching



Fingerprint
Recognition



Vein Patterns
Recognition



Signature
Recognition



Finger Geometry
Recognition



Retina Recognition



Face Recognition



Privacy Protection



Biometric
Recognition



Authentication



Biometric Data
Security



Iris Recognition

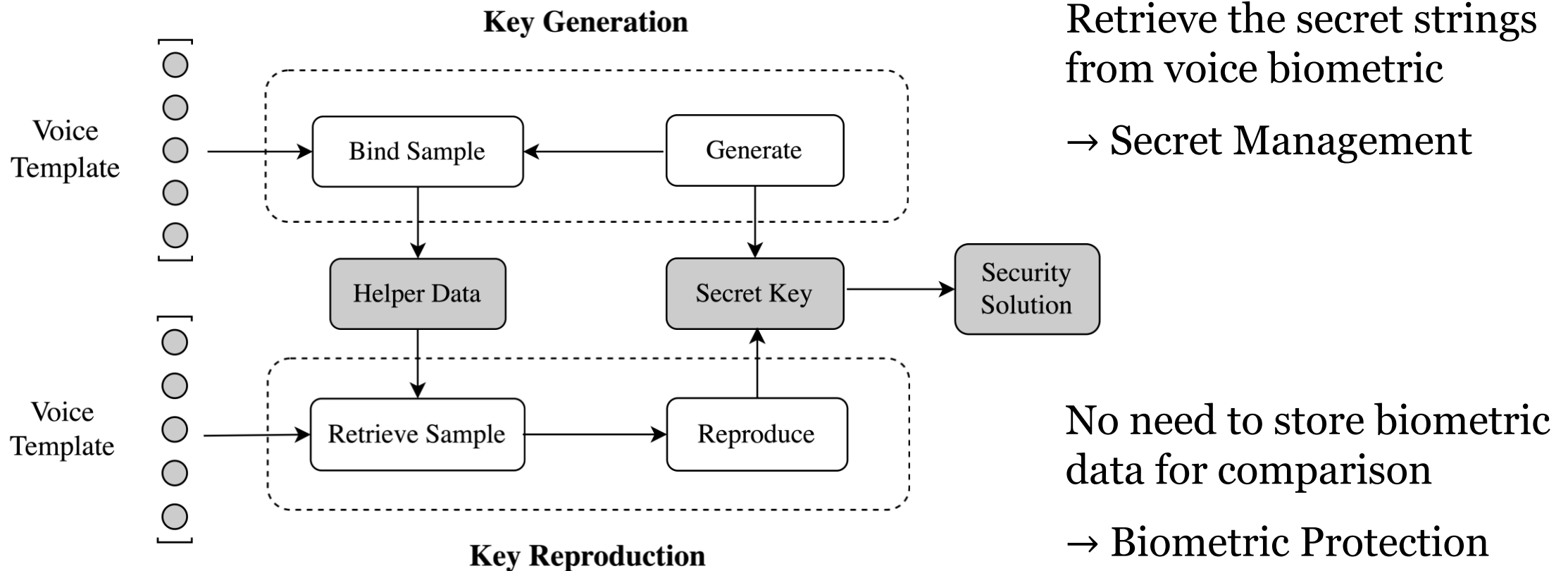


Getting Access



Hand Geometry
Recognition

Solution: Biometric Cryptosystem



Objectives

Proposed a new framework for the voice biometric cryptosystem.

Challenges

- *Correctness*: extract same secret if data is from the same person.
- *Secureness*: high entropy and uncorrelated secret for distinct users.
- *Revocability*: can revoke and regenerate a new secret strings.
- *Irreversibility*: can not reverse information from helper data and secrets.
- *Unlinkability*: unable to link different secrets to a specific user.

Approach: Sample-Then-Lock

- Continuously samples random subsets from a noisy **binary source**.
- Uses each **subset as a key** to encrypt a uniform secret string.

Approach: Digital Lockers

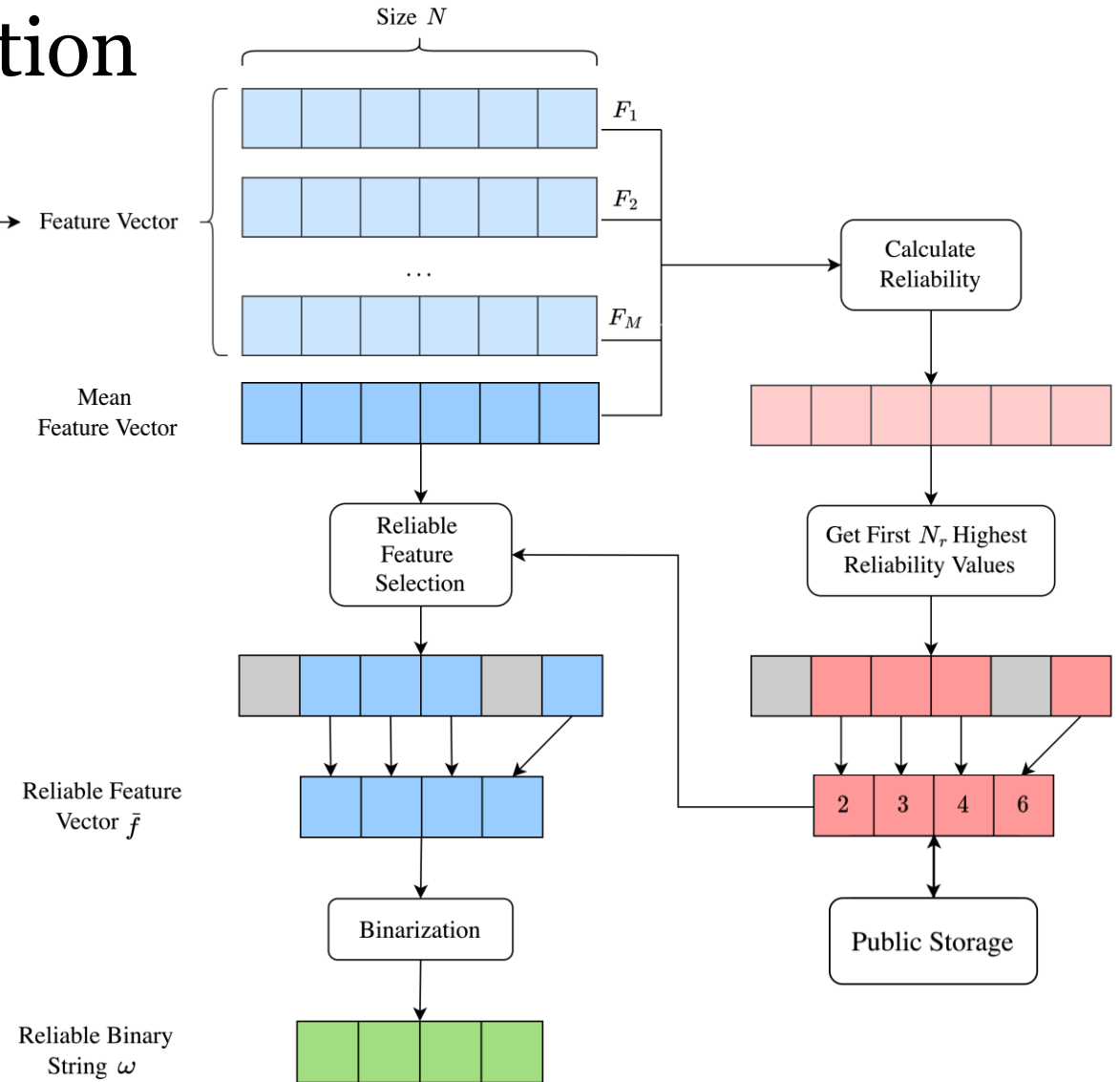
Encryption: $\text{lock}(\text{key}, \text{val}) = H(\text{nonce}, \text{key}) \oplus (\text{val} \parallel 0^s)$

- Concatenate with 0s $\rightarrow$ recognize wrong key with high probability.
- Hash nonce with key $\rightarrow$ correlation between similar keys disappears.

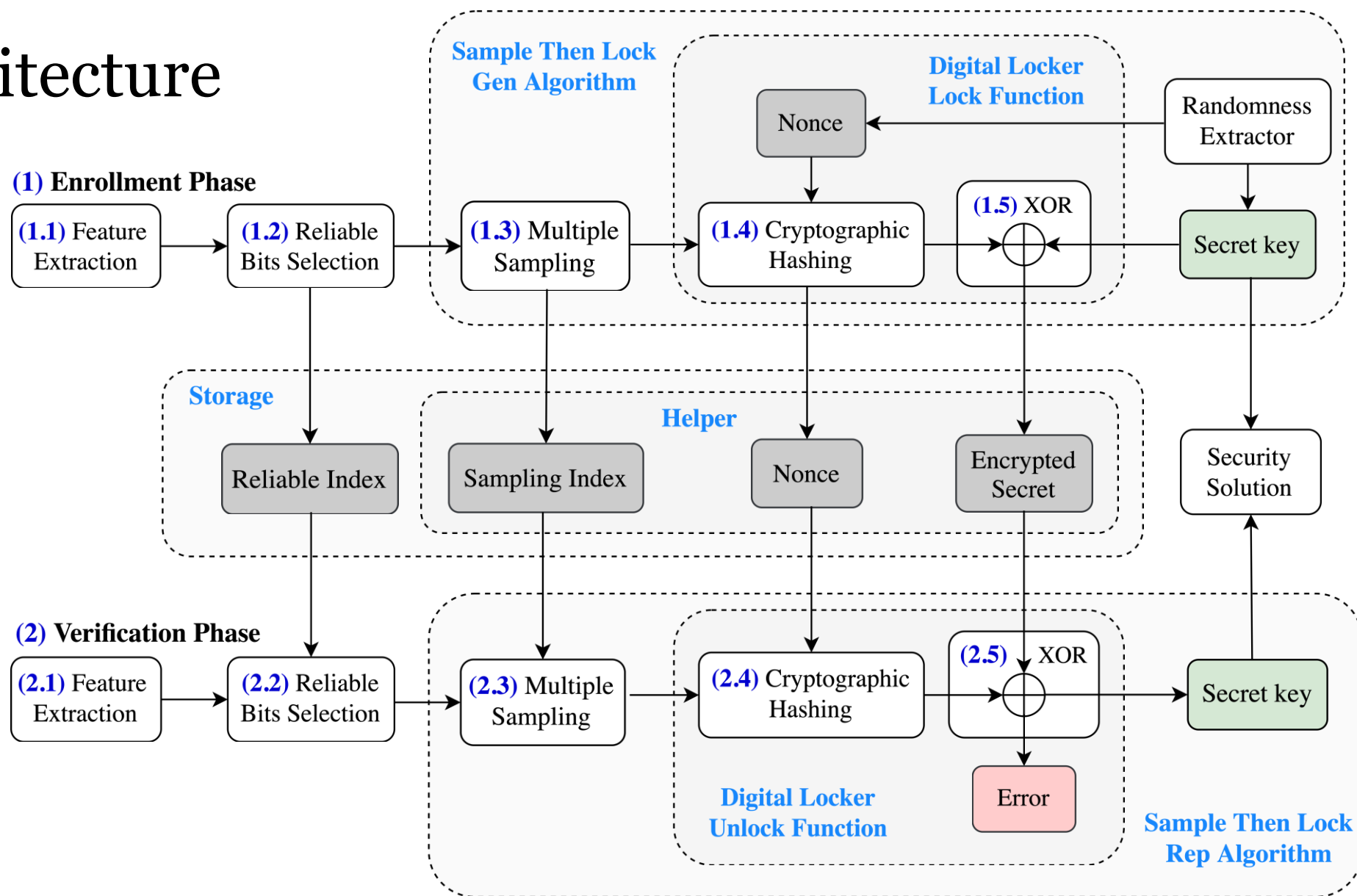
Approach: Reliable Bits Selection

$$\text{Reliability } r_i = \frac{-1}{M-1} \sum_{j=1}^M (\hat{f}_i^j - \bar{f}_i)^2$$

$$\text{Binarize } \omega_j = \text{sign}(\bar{f}_{i_j}) = \begin{cases} 1, & \text{if } \bar{f}_{i_j} \geq 0, \\ 0, & \text{if } \bar{f}_{i_j} < 0 \end{cases}$$



Architecture



Experimental Dataset (Preprocessed)

TIMIT (American English) for acoustic-phonetic studies

- 1386 vectors – 462 users. Each user has 3 vectors
- Enrollment 2 vectors – Verification 1 vector

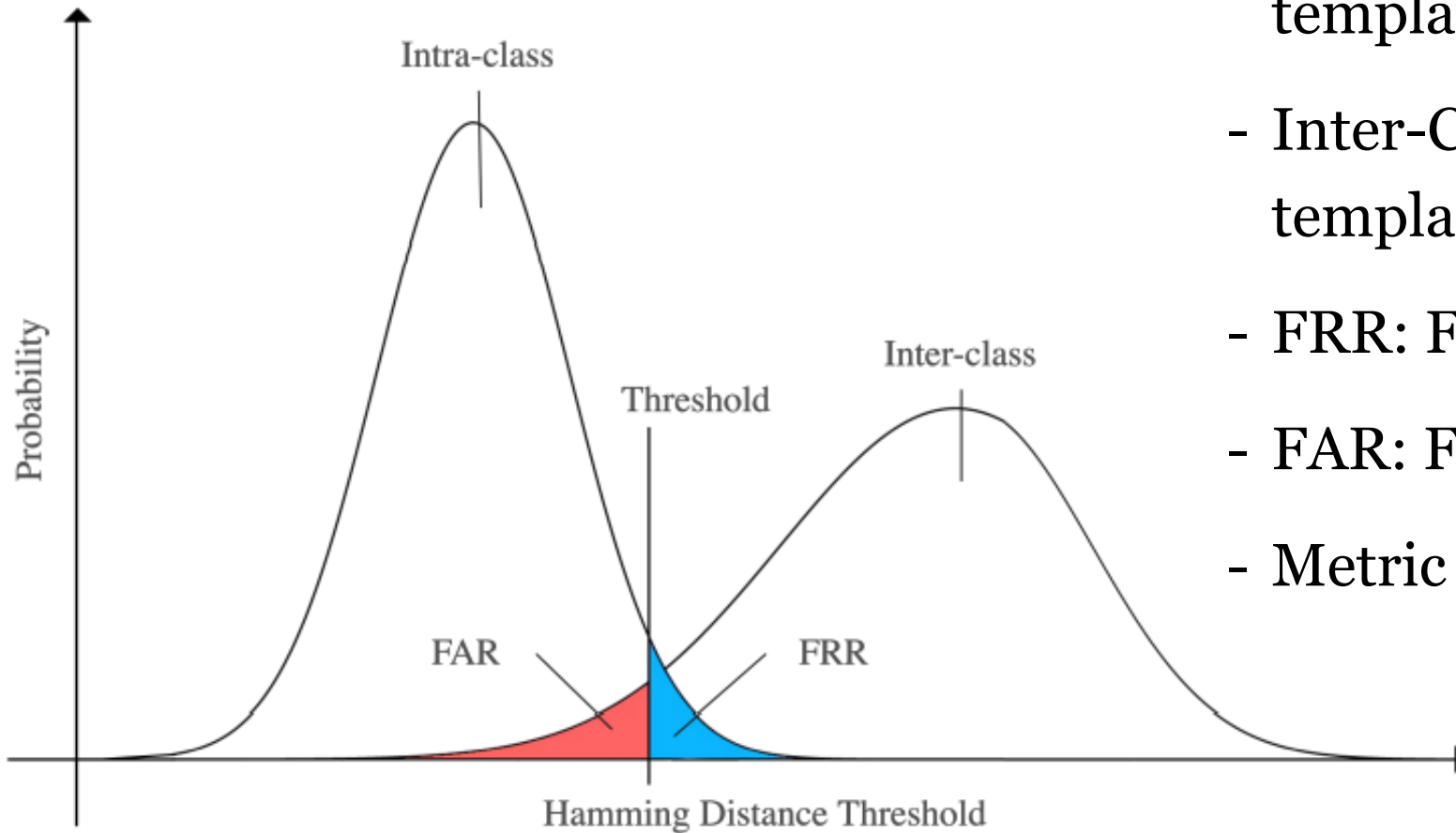
vKYC (Vietnamese) for speech command recognition

- 5154 vectors – 117 users.
- Enrollment 5 vectors – Verification 3 vectors

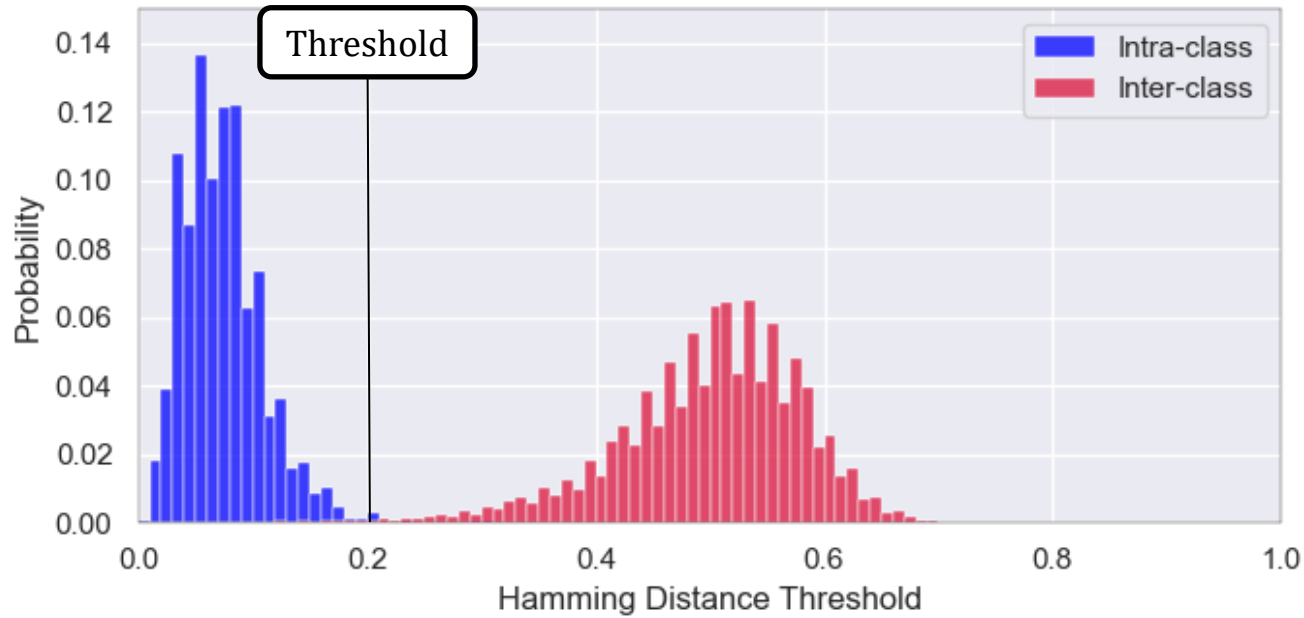
#Vectors/Speaker	1-9	10-19	20-39	40-69	70-99	100-149	≥ 150
#Speakers	36	3	24	29	13	8	4

Metric Measurement

- Intra-Class: all distances between templates from the same user
- Inter-Class: all distances between templates from distinct users
- FRR: False Rejection Rate
- FAR: False Acceptance Rate
- Metric Space: Hamming Distance



vKYC



Before Using Framework

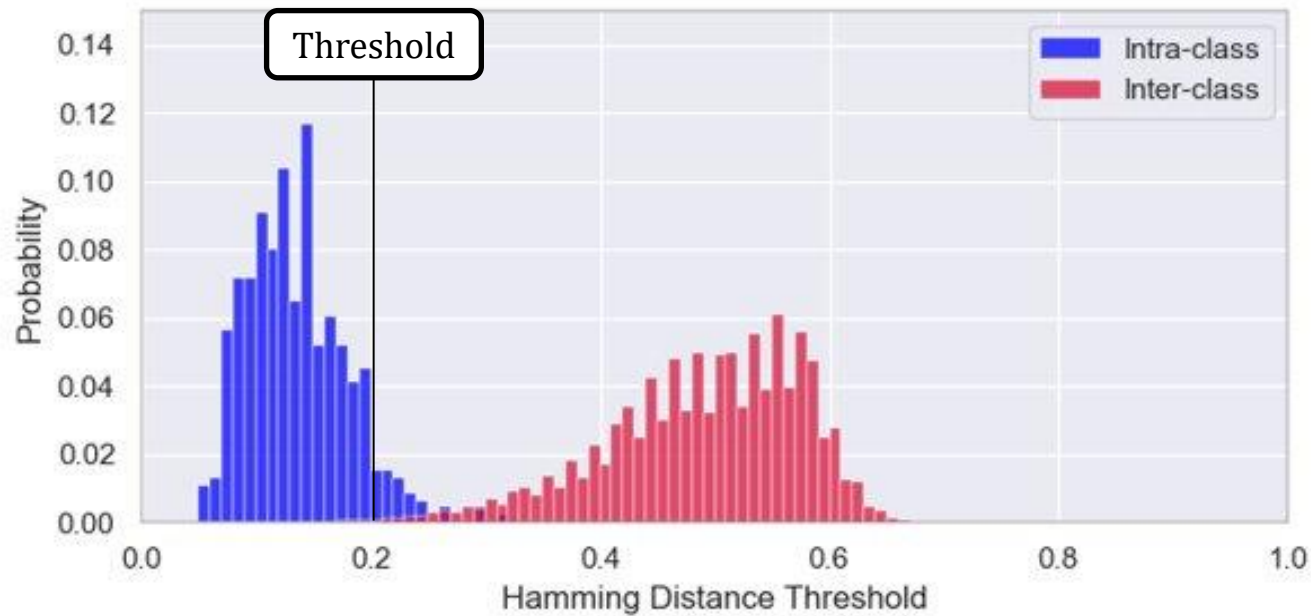
	TH	Tests	Cases/Test	Average
FRR	0.20	100	101	0.43%
FAR			13456	0.61%

After Using Framework

	TH	Cases	False Times	Percentage
FRR	0.20	1486	7	0.47%
FAR		192741	915	0.47%

	TH	Cases	False Times	Percentage
FRR	0.20	101	1	0.99%
FAR		13456	36	0.27%

TIMIT



Before Using Framework

	TH	Tests	Cases/Test	Average
FRR	0.20	100	462	8.35%
FAR			212982	0.23%

After Using Framework

	TH	Cases	False Times	Percentage
FRR	0.20	462	32	6.93%
FAR		638946	1451	0.23%

	TH	Cases	False Times	Percentage
FRR	0.20	462	47	10.17%
FAR		212982	47	0.022%

Security Analysis

- Correctness can be evaluated based on FRR
 - Security parameter: number of subsets (sampling times)
- Secureness can be first evaluated based on FAR
 - Secrets are long and uniform (generated from PRNG)
 - Based on security definition of Fuzzy Extractor + Digital Locker
- Revocability is based on reusability
 - Sample-Then-Lock is a reusable Fuzzy Extractor.

Security Analysis

- Irreversibility:
 - With secret key: Infeasible due to computational hardness of SHA-2
 - Without secret key: Infeasible based on the security of One Time Pad
- Unlinkability:
 - Secrets are uniform random between each enrollment (Revocability)
 - Linkability on input if reverse computation is feasible (Irreversibility)

Conclusion

- Proposed a new framework for voice biometric cryptosystem which can solve the problem of secret management and biometric protection.
- The difference in result before and after using our framework are around 2% for FRR and 0.4% for FAR on the feature vectors of vKYC and TIMIT.
- Analyzed the security properties based on building blocks of Digital Locker and Sample-Then-Lock.

Future Works

- Evaluate the entropy of binarized vectors to fully address the security.
- Experiment on other security assumptions: Decisional Diffie-Hellman, Learning With Error, Common Reference String
- Address the robustness property for active attacks and insider security.
- Reduce the size of helper data and computing time of enrollment phase for application on mobile devices.
- Further studies to generalize the framework on other biometrics.

Thank you for listening

Appendix

Definition [**Computational Fuzzy Extractor**]. Let $\mathcal{W}$ be a family of probability distributions over $\mathcal{M}$. A pair of randomized procedures “generate” (Gen) and “reproduce” (Rep) is an $(\mathcal{M}, \mathcal{W}, \kappa, t)$ -computational fuzzy extractor that is $(\epsilon_{sec}, s_{sec})$ -hard with error δ if Gen and Rep satisfy the following properties:

- The generate procedure Gen on input $w \in \mathcal{M}$ outputs an extracted string $r \in \{0, 1\}^\kappa$ and a helper string $p \in \{0, 1\}^*$.
- **Correctness** The reproduction procedure Rep takes an element $w' \in \mathcal{M}$ and a bit string $p \in \{0, 1\}^*$ as inputs. If $\text{dis}(w, w') \leq t$ and $(r, p) \leftarrow \text{Gen}(w)$, then $\Pr[\text{Rep}(w', p) = r] \geq 1 - \delta$, the probability is over the randomness of (Gen, Rep).
- **Security** For any distribution $W \in \mathcal{W}$, the extracted string r is pseudorandom conditioned on p , that is $\delta^{D_{s_{sec}}}((R, P), (U_\kappa, P)) \leq \epsilon_{sec}$

Definition [**Reusable Fuzzy Extractor**]. Given $\mathcal{W}$ as a distribution ensemble over $\mathcal{M}$. Let (Gen, Rep) be a $(\mathcal{M}, \mathcal{W}, \kappa, t)$ -computational fuzzy extractor that is $(\epsilon_{\text{sec}}, s_{\text{sec}})$ -hard with error δ . Let $(W^1, W^2, \dots, W^\rho)$ be ρ correlated random variables such that each W^j in $\mathcal{W}$. Let D be an adversary. Define the following game for all $j = 1, \dots, \rho$:

- **Sampling** The challenger sample $w^j \leftarrow W^j$ and $u \leftarrow \{0,1\}^\kappa$
- **Generation** The challenger computes $(r^j, p^j) \leftarrow \text{Gen}(w^j)$
- **Distinguishing** The advantage of D is

$$\begin{aligned} \text{Adv}(D) := & \Pr[D(r^1, \dots, r^{j-1}, r^j, r^{j+1}, \dots, r^\rho, p^1, \dots, p^\rho) = 1] \\ & - \Pr[D(r^1, \dots, r^{j-1}, u^j, r^{j+1}, \dots, r^\rho, p^1, \dots, p^\rho) = 1] \end{aligned}$$

(Gen, Rep) is $(\rho, \epsilon_{\text{sec}}, s_{\text{sec}})$ -reusable if for all $D \in \mathcal{D}_{\text{sec}}$ and all $j = 1, \dots, \rho$, the advantage $\text{Adv}(D)$ is at most ϵ_{sec}

Definition [**Digital Locker**]. The pair of algorithms (lock, unlock) with the security parameter λ is an ℓ -composable secure digital locker with error γ if the following:

- **Correctness** For all key and val, $\Pr[\text{unlock}(\text{key}, \text{lock}(\text{key}, \text{val})) = \text{val}] \geq 1 - \gamma$
Furthermore, for any $\text{key}' \neq \text{key}$, $\Pr[\text{unlock}(\text{key}', \text{lock}(\text{key}, \text{val})) = \perp] \geq 1 - \gamma$
- **Security** For every PPT adversary A and every positive polynomial p , there exists a (possibly inefficient) simulator S and a polynomial $q(\lambda)$ such that for any sufficiently large s , any polynomially-long sequence of values $(\text{val}_i, \text{key}_i)$ for $i = 1, \dots, \ell$ and any auxiliary input $z \in \{0, 1\}^*$,

$$\left| \Pr \left[A \left(z, \{\text{lock}(\text{key}_i, \text{val}_i)\}_{i=1}^{\ell} \right) = 1 \right] - \Pr \left[S \left(z, \{|\text{key}_i|, |\text{val}_i|\}_{i=1}^{\ell} \right) = 1 \right] \right| \leq \frac{1}{p(s)}$$

where S is allowed $q(\lambda)$ queries to the oracles $\{\text{idealUnlock}(\text{key}_i, \text{val}_i)\}_{i=1}^{\ell}$

Definition [**Sample-Then-Lock**]. Let $\mathcal{Z}$ be an alphabet and let $W = W_1, \dots, W_n$ be a source with α -entropy k -samples, where each W_j is over $\mathcal{Z}$. Let ℓ be a positive parameter, to be decided. Let $(\text{lock}, \text{unlock})$ be an ℓ -composable secure digital locker with error γ (for κ -bit keys and values over $\mathcal{Z}^k$). Define Gen, Rep as follows:

Algorithm 1: Gen

1. Input $w = w_1, \dots, w_n$
 2. Sample $r \xleftarrow{\$} \{0, 1\}^\kappa$
 3. For $i = 1, \dots, \ell$:
 - (i) Choose random $1 \leq j_{i,1}, \dots, j_{i,k} \leq n$
 - (ii) Set $v_i = w_{j_{i,1}}, \dots, w_{j_{i,k}}$
 - (iii) Set $c_i = \text{lock}(v_i, r)$
 - (iv) Set $p_i = c_i, (j_{i,1}, \dots, j_{i,k})$
 4. Output (r, p) where $p = p_1 \dots p_\ell$
-

Algorithm 2: Rep

1. Input $(w' = w'_1, \dots, w'_n, p = p_1 \dots p_n)$
 2. For $i = 1, \dots, \ell$:
 - (i) Parse $p_i = c_i, (j_{i,1}, \dots, j_{i,k})$
 - (ii) Set $v'_i = w'_{j_{i,1}}, \dots, w'_{j_{i,k}}$
 - (iii) Set $r_i = \text{unlock}(v'_i, c_i)$. If $r_i \neq \perp$ output r_i
 3. Output $\perp$
-

Security [**Sample-Then-Lock**]. Let λ be a security parameter and let $\mathcal{W}$ be a family of source over $\mathcal{Z}^n$ α -entropy k -samples for $\alpha = \log(\lambda)$. Let $(\text{lock}, \text{unlock})$ be an ℓ -composable secure digital locker with error γ for κ -bit keys and values over $\mathcal{Z}^k$. Then for $s_{\text{sec}} = \text{poly}(\lambda)$, there exists some $\epsilon_{\text{sec}} = \text{negl}(\lambda)$ such that sample-then-lock is a $(\mathcal{Z}^n, \mathcal{W}, \kappa, t)$ -computational fuzzy extractor that is $(\epsilon_{\text{sec}}, s_{\text{sec}})$ -hard with error δ

$$\delta^{D_{s_{\text{sec}}}}((R, P), (U_{\kappa}, P)) = |\mathbb{E}[D(R, P)] - \mathbb{E}[D(U, P)]| \leq \epsilon_{\text{sec}}$$

Source with High-Entropy Samples

Let the source $W = W_1, W_2, \dots, W_n$ consisting of strings of length n over some arbitrary alphabet $\mathcal{Z}$. The case of greatest interest is that of the binary alphabet $\mathcal{Z} = \{0, 1\}$. For some parameters k and α , the source W is said to be a source with α -entropy k -samples if $\tilde{H}_\infty(W_{j_1}, \dots, W_{j_k} \mid j_1, \dots, j_k) \geq \alpha$ for k uniformly random indices $1 \leq j_1, \dots, j_k \leq n$.

Sample-Then-Lock

- Ensure entropy on any subset size k
- Works on binary alphabets and is reusable
- Computational Fuzzy Extractor

Other constructions

- (Error-Correct-and-Obfuscate) Many symbols have high entropy individually
- (Condense-then-Fuzzy-Extract) Sufficient entropy conditioned between symbols
- Cannot work on small alphabets and is not reusable
- Computational Fuzzy Conductor (HILL Entropy)

Biometric System Security Requirements

- Confidentiality
- Integrity
- Availability
- Revocability

Biometric Information Privacy Requirements

- Irreversibility
- Unlinkability
- Confidentiality
- Data Minimization